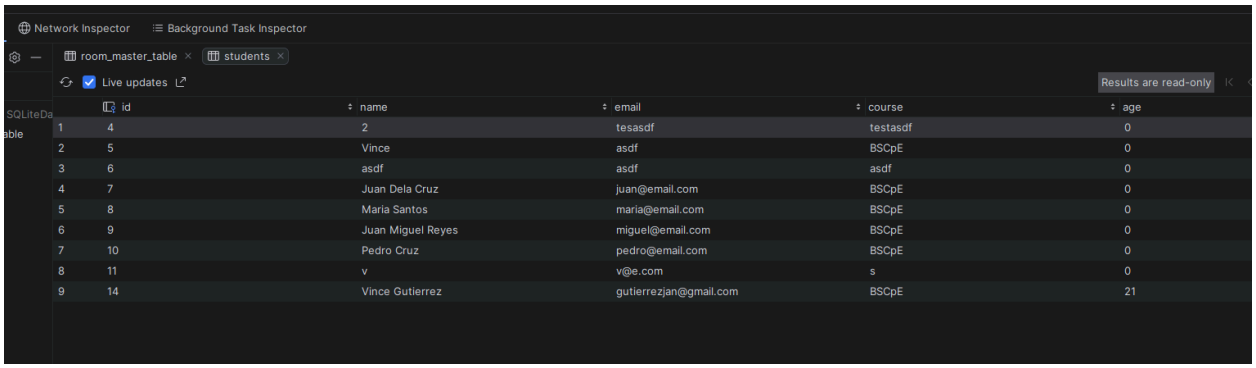
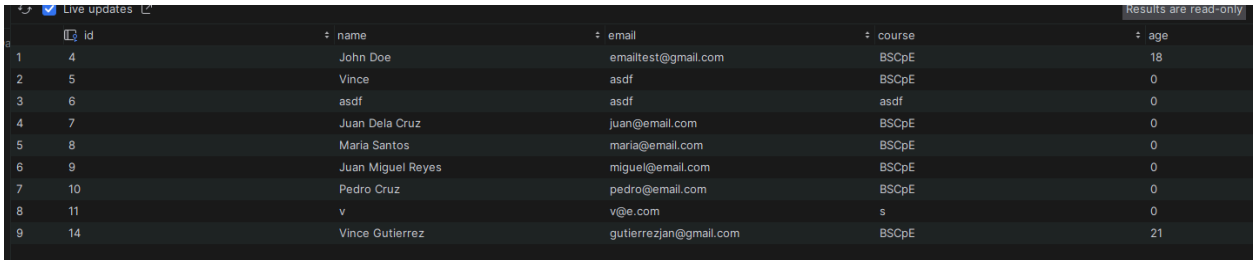
Gutierrez, Jan Vince Chrisean T.
BSCpE 4-A

Updating entry ID 4
2 - tesasdf - testasdf - 0



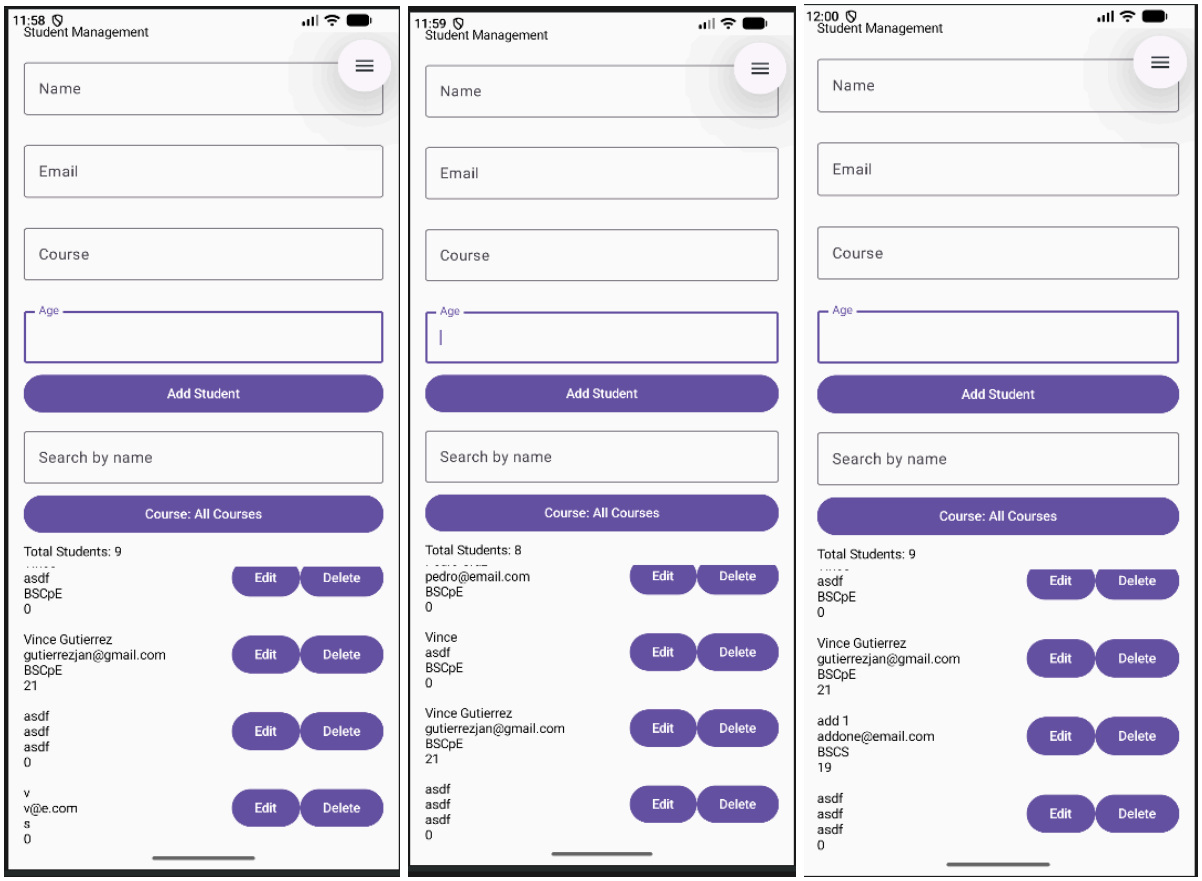
	id	name	email	course	age
1	4	2	tesasdf	testasdf	0
2	5	Vince	asdf	BSCpE	0
3	6	asdf	asdf	asdf	0
4	7	Juan Dela Cruz	juan@email.com	BSCpE	0
5	8	Maria Santos	maria@email.com	BSCpE	0
6	9	Juan Miguel Reyes	miguel@email.com	BSCpE	0
7	10	Pedro Cruz	pedro@email.com	BSCpE	0
8	11	v	v@e.com	s	0
9	14	Vince Gutierrez	gutierrezjan@gmail.com	BSCpE	21

John Doe - emailtest@gmail.com - BSCpE - 18

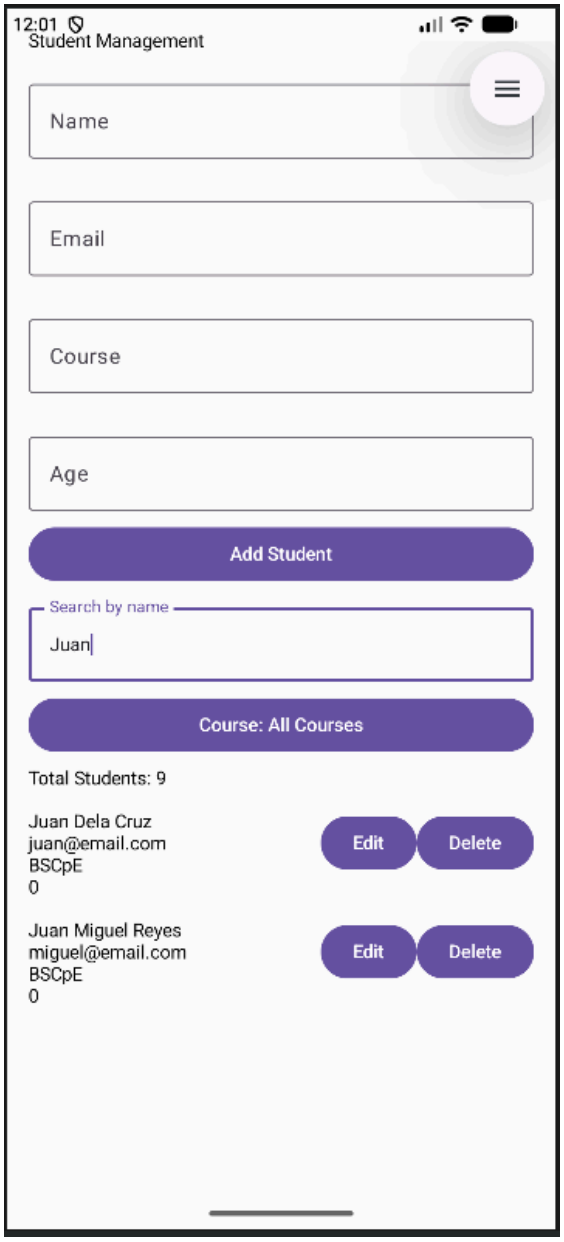


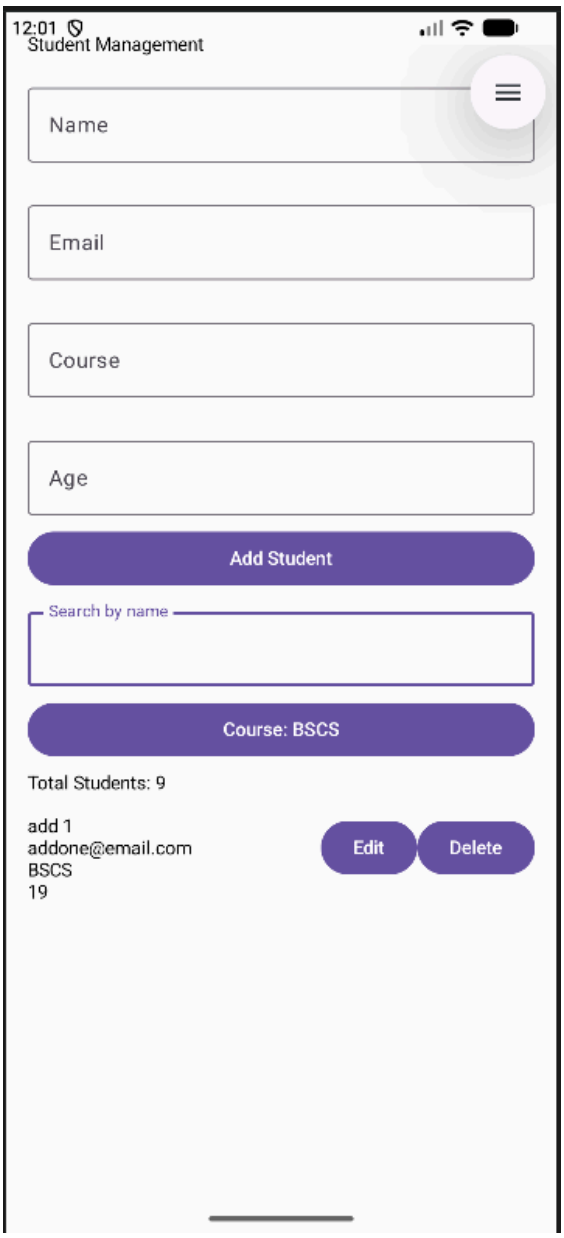
	id	name	email	course	age
1	4	John Doe	emailtest@gmail.com	BSCpE	18
2	5	Vince	asdf	BSCpE	0
3	6	asdf	asdf	asdf	0
4	7	Juan Dela Cruz	juan@email.com	BSCpE	0
5	8	Maria Santos	maria@email.com	BSCpE	0
6	9	Juan Miguel Reyes	miguel@email.com	BSCpE	0
7	10	Pedro Cruz	pedro@email.com	BSCpE	0
8	11	v	v@e.com	s	0
9	14	Vince Gutierrez	gutierrezjan@gmail.com	BSCpE	21

Reactive total number of students

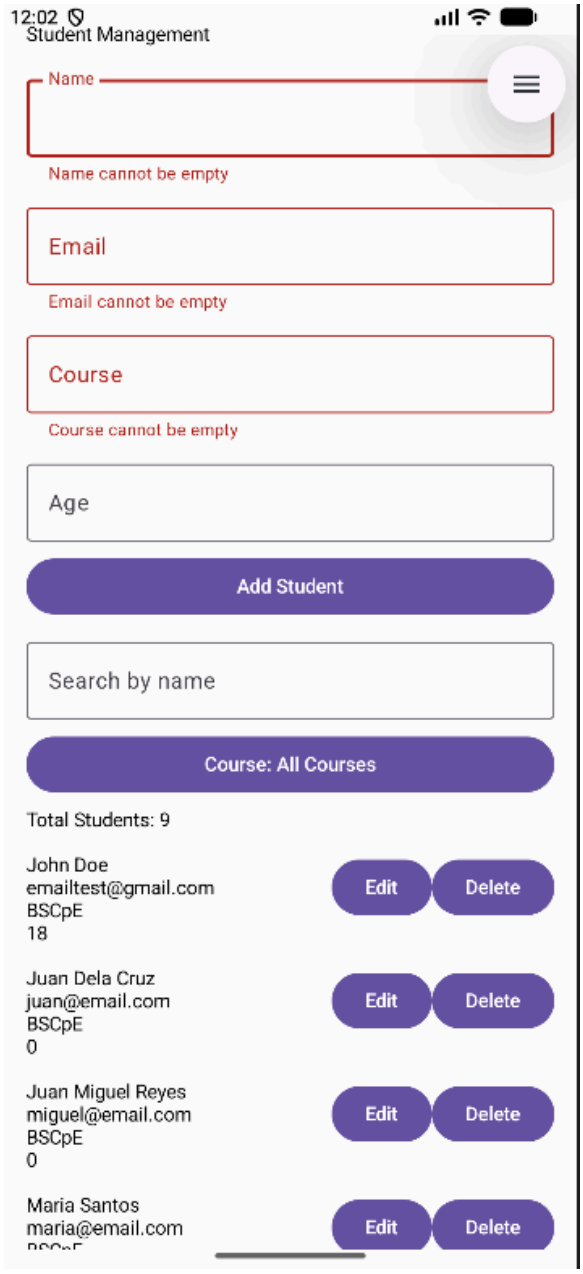


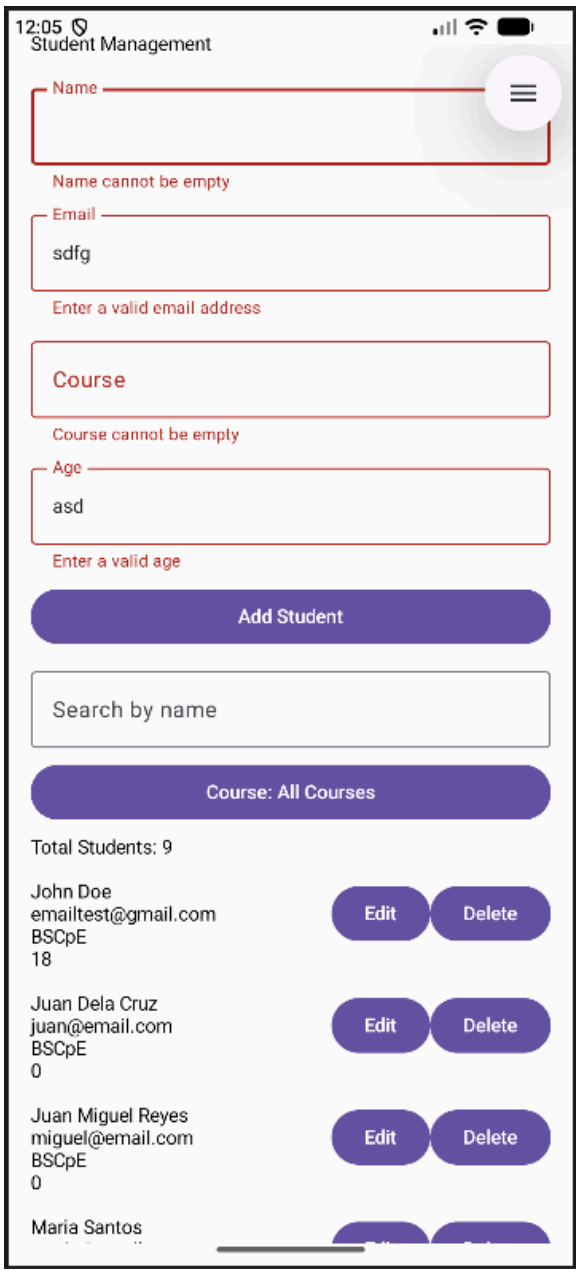
Name search and filtering





Input Validation





Database Migration

Note: All entries with age 0 were prior migration, i forgot to take ss of before and after po.

	id	name	email	course	age
1	4	John Doe	emailtest@gmail.com	BSCpE	18
2	5	Vince	asdf	BSCpE	0
3	6	asdf	asdf	asdf	0
4	7	Juan Dela Cruz	juan@email.com	BSCpE	0
5	8	Maria Santos	maria@email.com	BSCpE	0
6	9	Juan Miguel Reyes	miguel@email.com	BSCpE	0
7	10	Pedro Cruz	pedro@email.com	BSCpE	0
8	14	Vince Gutierrez	gutierrezjan@gmail.com	BSCpE	21
9	15	add 1	addone@email.com	BSCS	19

Why must the student retain the same primary key when updating an existing record?

The primary key identifies the exact database row to edit. Keeping the same ID lets Room update that student. A different ID could lead to unexpected behavior, while a missing one could create a new record instead.

What is the purpose of the SQL LIKE operator?

LIKE searches text by matching the same pattern or string of text. Example: '%Juan%'; the % symbols mean any characters before or after "Juan," so it finds names containing that text

Why does the DAO return Flow<List<Student>> instead of List<Student>?

Flow automatically emits a new list whenever it observes or detects database changes. This keeps the UI updated without the need for manual refreshing.

Why should the UI communicate with the ViewModel instead of calling the DAO directly?

The ViewModel separates UI code from data logic. This keeps the architecture organized, makes the UI simpler, and prevents direct database access from the UI. Essentially, the same principle of separating the layers in MVC.

What is the responsibility of the Repository in this implementation?

The Repository is the layer between the ViewModel and DAO. Its responsibility is to provide data operations to the ViewModel and keep the database access logic in one place.

Why should validation happen before calling the Repository?

Validation prevents invalid data from reaching the database. It also allows the app to show clear error messages to the user before saving. Essentially, validation happens before the repository call so that we ensure that entries going into the database are valid and not the other way around.

Why should validation be applied to both Add and Edit operations?

Both operations can save data; thus, we apply validation to both as they can alter or edit information in the database.

Why is a database-derived count more reliable than manually incrementing and decrementing a variable?

The database count is always reflective of the actual records. Manual counting becomes incorrect if an operation fails, wrong logic, or data changes from another part of the app.

How can Flow help keep the displayed count synchronized with database changes?

Room emits a new count through Flow whenever the students table changes. The ViewModel collects it, and the UI automatically displays the new updated total.

Why must the database version increase when the schema changes?

This is essentially versioning. Room uses the version number to detect changes. Increasing the number basically tells Room that there are changes and that it should run a migration before continuing operations.

What is a database migration?

A migration is code that changes an existing database schema, such as adding a column, while preserving already stored data. It's like renovating a house while people are still living in it.

Why is destructive migration inappropriate when existing student data must be preserved?

Destructive migration deletes and recreates the database, which removes all saved student records. This is inappropriate because we can not make use of destructive migration on important or sensitive records that make the system run. Also, in the activity, existing records are required to remain available.